

Comparison of IPv4 & IPv6

Internet Protocol (IP)

- IP is the underline protocol for all the other protocols in the TCP/IP protocol suite. IP defines the means to identify and reach a target computer on the network. Computer in the IP world are identify by unique numbers which are known as IP address.
- The Internet Protocol (IP) provides unreliable, connectionless packet delivery. IP is connectionless because it treats each packet of information independently.

IPv4

- **Internet Protocol version 4 (IPv4)** is the fourth revision in the development of the Internet Protocol (IP) and it is the first version of the protocol to be widely deployed.
- It is described in [IETF](#) publication [RFC 791](#) (September 1981) which rendered obsolete [RFC 760](#) (January 1980). The [United States Department of Defense](#) also standardized it as MIL-STD-1777.

- IPv4 is a data-oriented protocol to be used on a packet switched internetwork (e.g., Ethernet). It is a best effort delivery protocol in that it does not guarantee delivery, nor does it assure proper sequencing, or avoid duplicate delivery.

- IPv4 uses 32-bit (four-byte) addresses, which limits the address space to 4,294,967,296 (2^{32}) possible unique addresses.
- These aspects are addressed by an upper layer protocol (e.g. TCP, and partly by UDP).

Address representations

- IPv4 addresses are usually written in dot-decimal notation, which consists of the four octets of the address expressed in decimal and separated by periods. This is the base format used in the conversion in the following table:

Notation	Value	Conversion from dot-decimal
----------	-------	-----------------------------

IPv4 Header Format

Bits	0	3	4	7	9	15	16	31
Version	Header length		Type of service			Total length		
Identification						Flags	Fragment offset	
Time to live			Protocol			Header checksum		
32-bit source address								
32-bit destination address								
Options							Padding	

IPv4 Header Format

- **Version** :-The first header field in an IP packet is the four-bit version field. For IPv4, this has a value of 4 (hence the name IPv4).
- **Internet Header Length (IHL)** :-The second field is a four-bit Internet Header Length (IHL) telling the number of 32-bit words in the header.
- **Type of service** :-Contains five subfields that specify the precedence, delay, throughput, reliability, and cost desired for a packet. (The Internet does not guarantee this request.) This field is not widely used on the Internet.

IPv4 Header Format

- **Total length** :- The length of the datagram in bytes including the header, options, and the appended transport protocol segment or packet.
- **Identification** :- An integer that identifies the datagram.
- **Flags** :- Controls datagram fragmentation together with the identification field. The flags indicate whether the datagram may be fragmented, whether the datagram is fragmented, and whether the current fragment is the final one.
- **Fragment offset** :- The relative position of this fragment measured from the beginning of the original datagram in units of 8 bytes.

IPv4 Header Format

- **Time to live** :- How many routers a datagram can pass through. Each router decrements this value by 1 until it reaches 0 when the datagram is discarded. This keeps misrouted datagrams from remaining on the Internet forever.
- **Protocol** :-The high-level protocol type.
- **Header checksum**:- A number that is computed to ensure the integrity of the header values.
- **Source address** :- The 32-bit IPv4 address of the sending host.

IPv4 Header Format

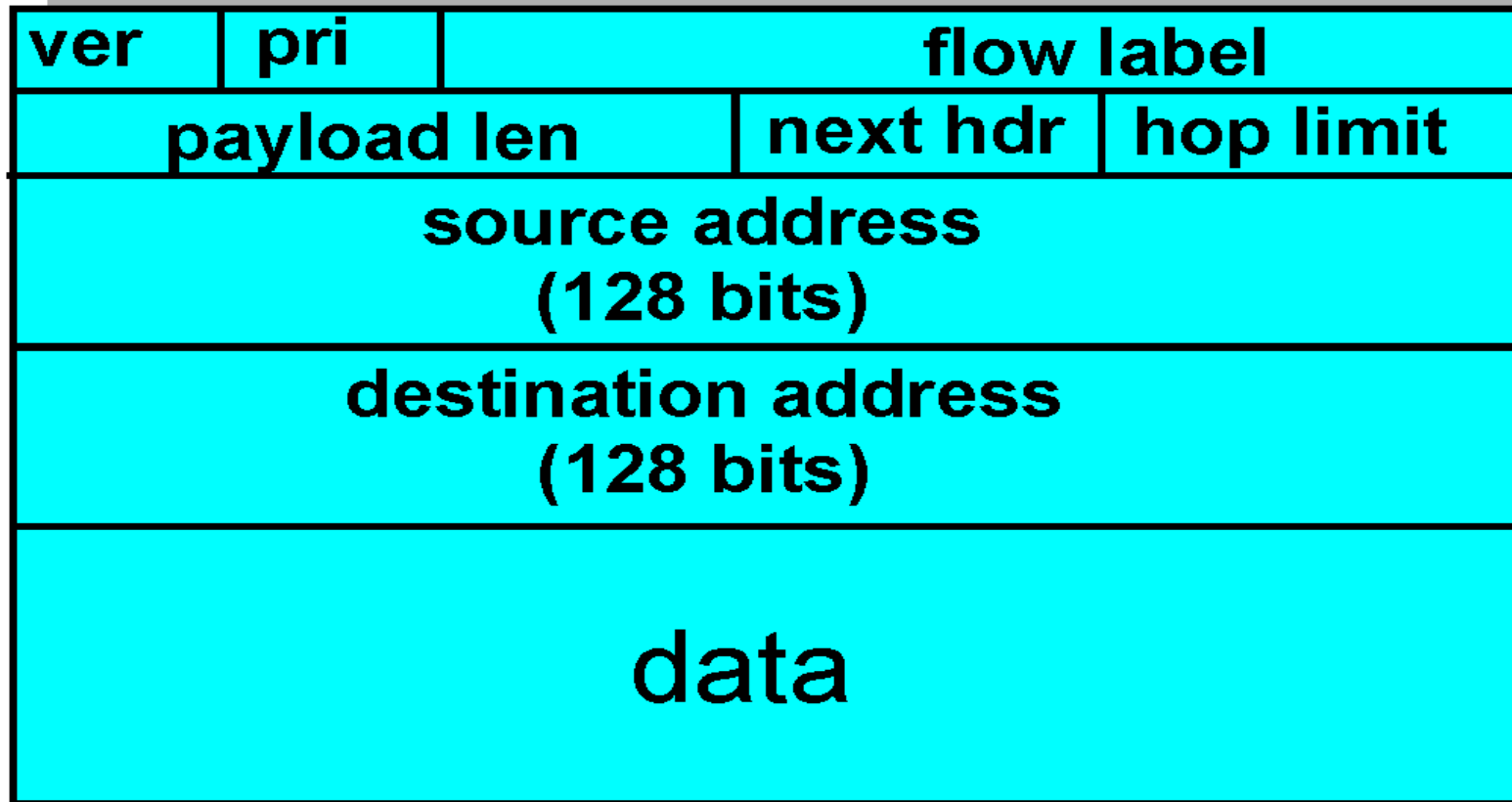
- **Destination address** :- The 32-bit IPv4 address of the receiving host.
- **Options** :- A list of optional specifications for security restrictions, route recording, and source routing. Not every datagram specifies an options field.
- **Padding** :- Null bytes which are added to make the header length an integral multiple of 32 bytes as required by the header length field.

IPv6

- The Internet Protocol version 6 (IPv6) effort is dynamic and is being driven by worldwide sites currently implementing and testing its functionality. Multiple Request for Comments (RFC) and Internet Drafts have been written to support those interested in joining the testing effort.
- Internet Protocol version 6 (IPv6) is the next generation of internet protocol, the current version is IPv4. The current exponential growth of the internet is the primary driver for the need of a new protocol.

- IPv6 supports most of the same functions and applications that were supported in IPv4. Those functions that were not successful in IPv4 were either not included or improved in IPv6.
- IPv6 include Ethernet, Point-to-Point Protocol (PPP), Fiber Distributed Data Interface (FDDI), Token Ring, Asynchronous Transfer Mode (ATM), Frame Relay, and IEEE 1394

IPv6 Header Format



← 32 bits →

IPv6 Header Format

- **Version:-** This four bit field identifies the IP version number. Not surprisingly, IPv6 carries a value of "6" in this field. Note that putting a "4" in this field does not create a valid IPv4 packet (if it did, life would be a lot simpler -- see the discussion below regarding the transition from IPv4 to IPv6).
- **Priority:-** This four bit field is similar in spirit to the ToS field we saw in IP version 4. [RFC 1883] states that values 0 through 7 are to be used for priority among traffic that is congestion-controlled (i.e., for which the source will back off on detection of congestion), while values 8 through 15 are used for non-congestion controlled traffic, such as constant bit rate real-time traffic.
- **Flow label:-** This field is used to identify a "flow" of packets.

IPv6 Header Format

- **Payload length**:- This 16-bit value is treated as an unsigned integer given the number of bytes in the IPv6 packet following the fixed length, 40 byte packet header.
- **Next header**:- This field identifies the protocol to which the contents (data field) of this packet will be delivered. The field uses the same values as the Protocol field in the IPv4 header.
- **Hop limit**:- The contents of this field are decremented by one by each router that forward the packet. If the hop limit count reaches zero, the packet is discarded.

IPv6 Header Format

- **Source and destination address:-** An IP v6 address has the following structure:



- **Data:-** This is the payload portion of the IPv6 packet. When the packet reaches its destination, the payload will be removed from the IP packet and passed on to the protocol specified in the next header field.

Comparison between IPv4 & IPv6

IPv4

- Source and destination addresses are 32 bits (4 bytes) in length.
- IPSec support is optional.
- IPv4 header does not identify packet flow for QoS handling by routers.
- Both routers and the sending host fragment packets.

IPv6

- Source and destination addresses are 128 bits (16 bytes) in length.
- IPSec support is required.
- IPv6 header contains Flow Label field, which identifies packet flow for QoS handling by router.
- Only the sending host fragments packets; routers do not.

Comparison between IPv4 & IPv6

IPv4

- Header includes a checksum.
- Header includes options.
- Address Resolution Protocol (ARP) uses broadcast ARP Request frames to resolve an IP address to a link-layer address.
- Internet Group Management Protocol (IGMP) manages membership in local subnet groups.

IPv6

- Header does not include a checksum.
- All optional data is moved to IPv6 extension headers.
- Multicast Neighbor Solicitation messages resolve IP addresses to link-layer addresses.
- Multicast Listener Discovery (MLD) messages manage membership in local subnet groups.

Comparison between IPv4 & IPv6

IPv4

- ICMP Router Discovery is used to determine the IPv4 address of the best default gateway, and it is optional.
- Broadcast addresses are used to send traffic to all nodes on a subnet.
- Must be configured either manually or through DHCP.

IPv6

- ICMPv6 Router Solicitation and Router Advertisement messages are used to determine the IP address of the best default gateway, and they are required.
- IPv6 uses a link-local scope all-nodes multicast address.
- Does not require manual configuration or DHCP.

Comparison between IPv4 & IPv6

IPv4

- Uses host address (A) resource records in Domain Name System (DNS) to map host names to IPv4 addresses.
- Uses pointer (PTR) resource records in the IN-ADDR.ARPA DNS domain to map IPv4 addresses to host names.
- Must support a 576-byte packet size (possibly fragmented).

IPv6

- Uses host address (AAAA) resource records in DNS to map host names to IPv6 addresses.
- Uses pointer (PTR) resource records in the IP6.ARPA DNS domain to map IPv6 addresses to host names.
- Must support a 1280-byte packet size (without fragmentation).

Transition Strategies

- Dual Stack
- Tunneling
- Header translation

References

- Data Communication & Networking by Behrouz A. Forouzan.
- www.wikipedia.org
- www.google.com
- <http://technet.microsoft.com/en-us/library/>

Thank you